42.8 SUPERCONDUCTIVITY

Superconductivity is the complete disappearance of all electrical resistance at low temperatures. We described this property at the end of Section 25.2 and the magnetic properties of type-I and type-II superconductors in Section 29.8. In this section we'll relate superconductivity to the structure and energy-band model of a solid.

Although superconductivity was discovered in 1911, it was not well understood on a theoretical basis until 1957. That year, the American physicists John Bardeen, Leon Cooper, and Robert Schrieffer published the theory of superconductivity, now called the BCS theory, that earned them the Nobel Prize in physics in 1972. (It was Bardeen's second; he shared his first for his work on the development of the transistor.) The key to the BCS theory is an interaction between *pairs* of conduction electrons, called *Cooper pairs,* caused by an interaction with the positive ions of the crystal. Here's a rough picture of what happens. A free electron exerts attractive forces on nearby positive ions, pulling them slightly closer together. The resulting slight concentration of positive charge then exerts an attractive force on another free electron with momentum opposite to the first. At ordinary temperatures this electron-pair interaction is very small in comparison to energies of thermal motion, but at very low temperatures it is significant.

Bound together this way, the pairs of electrons cannot *individually* gain or lose very small amounts of energy, as they would ordinarily be able to do in a partly filled conduction band. Their pairing gives an energy gap in the allowed electron quantum levels, and at low temperatures there is not enough collision energy to jump this gap. Therefore the electrons can move freely through the crystal without any energy exchange through collisions—that is, with zero resistance.

Since 1987 physicists have discovered a number of compounds that remain superconducting at temperatures above 77 K (the boiling point of liquid nitrogen). The original pairing mechanism of the BCS theory cannot explain the properties of these *high-temperature superconductors.* Instead, it appears that electrons in these materials form pairs due to magnetic interactions between their spins.

CHAPTER 42 SUMMARY

Molecular bonds and molecular spectra: The principal types of molecular bonds are ionic, covalent, van der Waals, and hydrogen bonds. In a diatomic molecule the rotational energy levels are given by Eq. (42.3), where I is the moment of inertia of the molecule, m_r is its reduced mass, and r_0 is the distance between the two atoms. The vibrational energy levels are given by Eq. (42.7), where k' is the effective force constant of the interatomic force. (See Examples 42.1–42.3.)

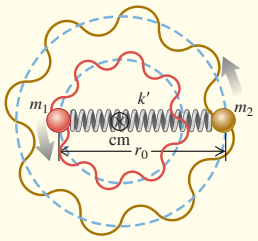
$$E_l = l(l+1)\frac{\hbar^2}{2I} \quad (l = 0, 1, 2, \ldots) \qquad (42.3)$$

$$I = m_r r_0^2 \qquad (42.6)$$

$$m_r = \frac{m_1 m_2}{m_1 + m_2} \qquad (42.4)$$

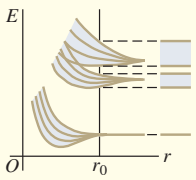
$$E_n = (n + \tfrac{1}{2})\hbar\omega = (n + \tfrac{1}{2})\hbar\sqrt{\frac{k'}{m_r}} \qquad (42.7)$$

$$(n = 0, 1, 2, \ldots)$$



Solids and energy bands: Interatomic bonds in solids are of the same types as in molecules plus one additional type, the metallic bond. Associating the basis with each lattice point gives the crystal structure. (See Example 42.4.)

When atoms are bound together in condensed matter, their outer energy levels spread out into bands. At absolute zero, insulators and conductors have a completely filled valence band separated by an energy gap from an empty conduction band. Conductors, including metals, have partially filled conduction bands. (See Example 42.5.)

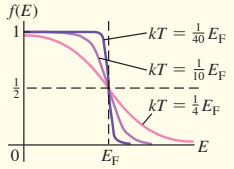


Continued

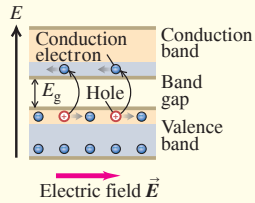
Free-electron model of metals: In the free-electron model of the behavior of conductors, the electrons are treated as completely free particles within the conductor. In this model the density of states is given by Eq. (42.15). The probability that an energy state of energy E is occupied is given by the Fermi–Dirac distribution, Eq. (42.16), which is a consequence of the exclusion principle. In Eq. (42.16), E_F is the Fermi energy. (See Examples 42.6–42.8.)

$$g(E) = \frac{(2m)^{3/2}V}{2\pi^2\hbar^3}E^{1/2} \qquad (42.15)$$

$$f(E) = \frac{1}{e^{(E-E_F)/kT} + 1} \qquad (42.16)$$

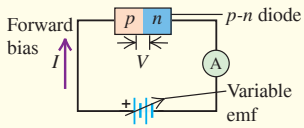


Semiconductors: A semiconductor has an energy gap of about 1 eV between its valence and conduction bands. Its electrical properties may be drastically changed by the addition of small concentrations of donor impurities, giving an n-type semiconductor, or acceptor impurities, giving a p-type semiconductor. (See Example 42.9.)



Semiconductor devices: Many semiconductor devices, including diodes, transistors, and integrated circuits, use one or more p-n junctions. The current–voltage relationship for an ideal p-n junction diode is given by Eq. (42.22).

$$I = I_S(e^{eV/kT} - 1) \qquad (42.22)$$



Chapter 42 Media Assets

GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 42.2 and 42.3 (Section 42.2) before attempting these problems.

VP42.3.1 The two nuclei in the nitric oxide (NO) molecule are 0.1154 nm apart. The mass of the most common nitrogen atom is 2.326×10^{-26} kg, and the mass of the most common oxygen atom is 2.656×10^{-26} kg. Find (a) the reduced mass of the NO molecule, (b) the moment of inertia of the NO molecule, and (c) the energies, in meV, of the lowest three rotational energy levels of NO.

VP42.3.2 The $l = 2$ rotational level of the hydrogen chloride (HCl) molecule has energy 7.90 meV. The mass of the most common hydrogen atom is 1.674×10^{-27} kg, and the mass of the most common chlorine atom is 5.807×10^{-26} kg. Find (a) the moment of inertia of the HCl molecule, (b) the reduced mass of the HCl molecule, and (c) the distance between the H and Cl nuclei.

VP42.3.3 The mass of the most common silicon atom is 4.646×10^{-26} kg, and the mass of the most common oxygen atom is 2.656×10^{-26} kg. When a molecule of silicon monoxide (SiO) makes a transition between the $l = 1$ and $l = 0$ rotational levels, it emits a photon of wavelength 6.882 mm. Find (a) the moment of inertia of the SiO molecule, (b) the reduced mass of the SiO molecule, and (c) the distance between the Si and O nuclei.

VP42.3.4 A CO molecule is initially in the $n = 2$ vibrational level. If this molecule loses both vibrational and rotational energy and emits a photon, what are the photon wavelength and frequency if the initial angular momentum quantum number is (a) $l = 3$ and (b) $l = 2$?

Be sure to review EXAMPLES 42.6 and 42.7 (Section 42.5) before attempting these problems.

VP42.7.1 For free electrons in a solid with Fermi energy E_F and temperature T, find the energy for which the probability that a state at that energy is occupied is (a) 0.33 and (b) 0.90.

VP42.7.2 The Boltzmann constant is $k = 8.617 \times 10^{-5}$ eV/K. For a metallic solid at room temperature (293 K), what is the probability that an electron state is occupied if its energy is (a) 0.0250 eV below the Fermi level, (b) 0.0400 eV above the Fermi level, and (c) 0.100 eV above the Fermi level?

VP42.7.3 Silver contains 5.8×10^{28} free electrons per cubic meter. At absolute zero, what are (a) the Fermi energy (in J and eV) of silver, (b) the speed of an electron with this energy, and (c) the density of states (in states/J and states/eV) at the Fermi energy for a block of silver of volume 1.0 cm^3?

VP42.7.4 If kT is small compared to the Fermi energy at absolute zero, the Fermi energy at temperature T is essentially the same as at absolute zero. Use the results of the previous problem to find the electron energy, in eV, for which the probability that an electron state is occupied is (a) 0.92 and (b) 1.0×10^{-4} for silver at room temperature (293 K).

Be sure to review EXAMPLE 42.9 (Section 42.6) before attempting these problems.

VP42.9.1 For a certain semiconductor, the Fermi energy is in the middle of its band gap. If the temperature of the semiconductor is 285 K, find the probability that a state at the bottom of the conduction band is occupied if the band gap is (a) 0.500 eV and (b) 1.50 eV.

VP42.9.2 Find the ratio of the probability that a state at the bottom of the conduction band of a semiconductor is occupied at 315 K to the probability at 295 K if the band gap is (a) 0.400 eV and (b) 0.800 eV. Assume that the Fermi energy is in the middle of the band gap.

VP42.9.3 Consider a material whose Fermi energy is in the middle of its band gap. What band-gap width provides a probability of 0.00190 that a state at the bottom of the conduction band is occupied if the temperature of the material is (a) 305 K and (b) 325 K?

VP42.9.4 In a certain semiconductor, the Fermi energy lies above the top of the valence band by an amount equal to $\frac{3}{4}$ of the band gap. Find the probability that a state at the bottom of the conduction band is occupied if the band gap is 0.300 eV and the temperature is (a) 275 K and (b) 325 K.

BRIDGING PROBLEM Molecular Vibration and Semiconductor Band Gap

At 80 K, the band gap in the semiconductor indium antimonide (InSb) is 0.230 eV. A photon emitted by a hydrogen fluoride (HF) molecule undergoing a vibration-rotation transition from $(n = 1, l = 0)$ to $(n = 0, l = 1)$ is absorbed by an electron at the top of the valence band of InSb. (a) How far above the top of the band gap (in eV) is the final state of the electron? (b) What is the probability that the final state was already occupied? The vibration frequency for HF is 1.24×10^{14} Hz, the mass of a hydrogen atom is 1.67×10^{-27} kg, the mass of a fluorine atom is 3.15×10^{-26} kg, and the equilibrium distance between the two nuclei is 0.092 nm. Assume that the Fermi energy for InSb is in the middle of the gap.

SOLUTION GUIDE

IDENTIFY and SET UP

1. This problem involves what you learned about molecular transitions in Section 42.2, about the Fermi–Dirac distribution in Section 42.5, and about semiconductors in Section 42.6.
2. Equation (42.9) gives the combined vibrational-rotational energy in the initial and final molecular states. The difference between the initial and final molecular energies equals the energy E of the emitted photon, which is in turn equal to the energy gained by the InSb valence electron when it absorbs that photon. The probability that the final state is occupied is given by the Fermi–Dirac distribution, Eq. (42.16).

EXECUTE

3. Before you can use Eq. (42.9), you'll first need to use the data given to calculate the moment of inertia I and the quantity $\hbar\omega$ for the HF molecule. (*Hint:* Be careful not to confuse frequency f and angular frequency ω.)
4. Use your results from step 3 to calculate the initial and final energies of the HF molecule. (*Hint:* Does the vibrational energy increase or decrease? What about the rotational energy?)
5. Use your result from step 4 to find the energy imparted to the InSb electron. Determine the final energy of this electron relative to the bottom of the conduction band.
6. Use your result from step 5 to determine the probability that the InSb final state is already occupied.

EVALUATE

7. Is the molecular transition of the HF molecule allowed? Which is larger: the vibrational energy change or the rotational energy change?
8. Is it likely that the excited InSb electron will be blocked from entering a state in the conduction band?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q42.1 Van der Waals bonds occur in many molecules, but hydrogen bonds occur only with materials that contain hydrogen. Why is this type of bond unique to hydrogen?

Q42.2 The bonding of gallium arsenide (GaAs) is said to be 31% ionic and 69% covalent. Explain.

Q42.3 The H_2^+ molecule consists of two hydrogen nuclei and a single electron. What kind of molecular bond do you think holds this molecule together? Explain.

Q42.4 The moment of inertia for an axis through the center of mass of a diatomic molecule calculated from the wavelength emitted in an $l = 19 \rightarrow l = 18$ transition is different from the moment of inertia calculated from the wavelength of the photon emitted in an $l = 1 \rightarrow l = 0$ transition. Explain this difference. Which transition corresponds to the larger moment of inertia?

Q42.5 Analysis of the photon absorption spectrum of a diatomic molecule shows that the vibrational energy levels for small values of n are very nearly equally spaced but the levels for large n are not equally spaced. Discuss the reason for this observation. Do you expect the adjacent levels to move closer together or farther apart as n increases? Explain.

Q42.6 Discuss the differences between the rotational and vibrational energy levels of the deuterium ("heavy hydrogen") molecule D_2 and those of the ordinary hydrogen molecule H_2. A deuterium atom has twice the mass of an ordinary hydrogen atom.

Q42.7 Various organic molecules have been discovered in interstellar space. Why were these discoveries made with radio telescopes rather than optical telescopes?

Q42.8 The air you are breathing contains primarily nitrogen (N_2) and oxygen (O_2). Many of these molecules are in excited rotational energy levels $(l = 1, 2, 3, \ldots)$, but almost all of them are in the vibrational ground level $(n = 0)$. Explain this difference between the rotational and vibrational behaviors of the molecules.

Q42.9 In what ways do atoms in a diatomic molecule behave as though they were held together by a spring? In what ways is this a poor description of the interaction between the atoms?

Q42.10 Individual atoms have discrete energy levels, but certain solids (which are made up of only individual atoms) show energy bands and gaps. What causes the solids to behave so differently from the atoms of which they are composed?

Q42.11 What factors determine whether a material is a conductor of electricity or an insulator? Explain.

Q42.12 Ionic crystals are often transparent, whereas metallic crystals are always opaque. Why?

Q42.13 Speeds of molecules in a gas vary with temperature, whereas speeds of electrons in the conduction band of a metal are nearly independent of temperature. Why are these behaviors so different?

Q42.14 Use the band model to explain how it is possible for some materials to undergo a semiconductor-to-metal transition as the temperature or pressure varies.

Q42.15 An isolated zinc atom has a ground-state electron configuration of filled $1s$, $2s$, $2p$, $3s$, $3p$, and $4s$ subshells. How can zinc be a conductor if its valence subshell is full?

Q42.16 The assumptions of the *free-electron model* of metals may seem contrary to reason, since electrons exert powerful electric forces on each other. Give some reasons why these assumptions actually make physical sense.

Q42.17 Why are materials that are good thermal conductors also good electrical conductors? What kinds of problems does this pose for the design of appliances such as clothes irons and electric heaters? Are there materials that do not follow this general rule?

Q42.18 What is the essential characteristic for an element to serve as a donor impurity in a semiconductor such as Si or Ge? For it to serve as an acceptor impurity? Explain.

Q42.19 There are several methods for removing electrons from the surface of a semiconductor. Can holes be removed from the surface? Explain.

Q42.20 A student asserts that silicon and germanium become good insulators at very low temperatures and good conductors at very high temperatures. Do you agree? Explain your reasoning.

Q42.21 The electrical conductivities of most metals decrease gradually with increasing temperature, but the intrinsic conductivity of semiconductors always *increases* rapidly with increasing temperature. What causes the difference?

Q42.22 How could you make compensated silicon that has twice as many acceptors as donors?

Q42.23 The saturation current I_S for a p-n junction, Eq. (42.22), depends strongly on temperature. Explain why.

Q42.24 Why does tunneling limit the miniaturization of MOSFETs?

EXERCISES

Section 42.1 Types of Molecular Bonds

42.1 • **An Ionic Bond.** (a) Calculate the electric potential energy for a K^+ ion and a Br^- ion separated by a distance of 0.29 nm, the equilibrium separation in the KBr molecule. Treat the ions as point charges. (b) The ionization energy of the potassium atom is 4.3 eV. Atomic bromine has an electron affinity of 3.5 eV. Use these data and the results of part (a) to estimate the binding energy of the KBr molecule. Do you expect the actual binding energy to be higher or lower than your estimate? Explain your reasoning.

42.2 • During each of these processes, a photon of light is given up. In each process, what wavelength of light is given up, and in what part of the electromagnetic spectrum is that wavelength? (a) A molecule decreases its vibrational energy by 0.198 eV; (b) an atom decreases its energy by 7.80 eV; (c) a molecule decreases its rotational energy by 4.80×10^{-3} eV.

42.3 • For the H_2 molecule the equilibrium spacing of the two protons is 0.074 nm. The mass of a hydrogen atom is 1.67×10^{-27} kg. Calculate the wavelength of the photon emitted in the rotational transition $l = 3$ to $l = 1$.

Section 42.2 Molecular Spectra

42.4 •• The H_2 molecule has a moment of inertia of $4.6 \times 10^{-48}\ \text{kg}\cdot\text{m}^2$. What is the wavelength λ of the photon absorbed when H_2 makes a transition from the $l = 3$ to the $l = 4$ rotational level?

42.5 • A hypothetical NH molecule makes a rotational-level transition from $l = 3$ to $l = 1$ and gives off a photon of wavelength 1.710 nm in doing so. What is the separation between the two atoms in this molecule if we model them as point masses? The mass of hydrogen is 1.67×10^{-27} kg, and the mass of nitrogen is 2.33×10^{-26} kg.

42.6 • Two atoms of cesium (Cs) can form a Cs_2 molecule. The equilibrium distance between the nuclei in a Cs_2 molecule is 0.447 nm. Calculate the moment of inertia about an axis through the center of mass of the two nuclei and perpendicular to the line joining them. The mass of a cesium atom is 2.21×10^{-25} kg.

42.7 •• **CP** The rotational energy levels of CO are calculated in Example 42.2. If the energy of the rotating molecule is described by the classical expression $K = \frac{1}{2}I\omega^2$, for the $l = 1$ level what are (a) the angular speed of the rotating molecule; (b) the linear speed of each atom; (c) the rotational period (the time for one rotation)?

42.8 • The vibrational and rotational energies of the CO molecule are given by Eq. (42.9). Calculate the wavelength of the photon absorbed by CO in each of these vibration-rotation transitions: (a) $n = 0$, $l = 2 \rightarrow n = 1$, $l = 3$; (b) $n = 0$, $l = 3 \rightarrow n = 1$, $l = 2$; (c) $n = 0$, $l = 4 \rightarrow n = 1$, $l = 3$.

42.9 • A lithium atom has mass 1.17×10^{-26} kg, and a hydrogen atom has mass 1.67×10^{-27} kg. The equilibrium separation between the two nuclei in the LiH molecule is 0.159 nm. (a) What is the difference in energy between the $l = 3$ and $l = 4$ rotational levels? (b) What is the wavelength of the photon emitted in a transition from the $l = 4$ to the $l = 3$ level?

42.10 • If a sodium chloride (NaCl) molecule could undergo an $n \rightarrow n - 1$ vibrational transition with no change in rotational quantum number, a photon with wavelength 20.0 μm would be emitted. The mass of a sodium atom is 3.82×10^{-26} kg, and the mass of a chlorine atom is 5.81×10^{-26} kg. Calculate the force constant k' for the interatomic force in NaCl.

42.11 •• When a hypothetical diatomic molecule having atoms 0.8900 nm apart undergoes a rotational transition from the $l = 2$ state to the next lower state, it gives up a photon having energy 8.800×10^{-4} eV. When the molecule undergoes a vibrational transition from one energy state to the next lower energy state, it gives up 0.2590 eV. Find the force constant of this molecule.

Section 42.3 Structure of Solids

42.12 • Potassium bromide (KBr) has a density of $2.75 \times 10^3\ \text{kg/m}^3$ and the same crystal structure as NaCl. The mass of a potassium atom is 6.49×10^{-26} kg, and the mass of a bromine atom is 1.33×10^{-25} kg. (a) Calculate the average spacing between adjacent atoms in a KBr crystal. (b) How does the value calculated in part (a) compare with the spacing in NaCl (see Exercise 42.13)? Is the relationship between the two values qualitatively what you would expect? Explain.

42.13 • **Density of NaCl**. The spacing of adjacent atoms in a crystal of sodium chloride is 0.282 nm. The mass of a sodium atom is 3.82×10^{-26} kg, and the mass of a chlorine atom is 5.89×10^{-26} kg. Calculate the density of sodium chloride.

Section 42.4 Energy Bands

42.14 • The gap between valence and conduction bands in diamond is 5.47 eV. (a) What is the maximum wavelength of a photon that can excite an electron from the top of the valence band into the conduction band? In what region of the electromagnetic spectrum does this photon lie? (b) Explain why pure diamond is transparent and colorless. (c) Most gem diamonds have a yellow color. Explain how impurities in the diamond can cause this color.

42.15 • The maximum wavelength of light that a certain silicon photocell can detect is 1.11 μm. (a) What is the energy gap (in electron volts) between the valence and conduction bands for this photocell? (b) Explain why pure silicon is opaque.

42.16 • The gap between valence and conduction bands in silicon is 1.12 eV. A nickel nucleus in an excited state emits a gamma-ray photon with wavelength 9.31×10^{-4} nm. How many electrons can be excited from the top of the valence band to the bottom of the conduction band by the absorption of this gamma ray?

Section 42.5 Free-Electron Model of Metals

42.17 • Calculate the density of states $g(E)$ for the free-electron model of a metal if $E = 7.8$ eV and $V = 1.0\ \text{cm}^3$. Express your answer in units of states per electron volt.

42.18 • The Fermi energy of sodium is 3.23 eV. (a) Find the average energy E_{av} of the electrons at absolute zero. (b) What is the speed of an electron that has energy E_{av}? (c) At what Kelvin temperature T is kT equal to E_F? (This is called the *Fermi temperature* for the metal. It is approximately the temperature at which molecules in a classical ideal gas would have the same kinetic energy as the fastest-moving electron in the metal.)

42.19 • Consider the density of states in the free-electron model for an energy of 1.60 eV. At what energy will the density of states be (a) doubled and (b) halved?

42.20 • For a metal with kT equal to 25% of the Fermi energy, what is the probability that a state will be occupied by an electron if its energy is (a) 50% of the Fermi energy and (b) 50% greater than the Fermi energy? Are your results consistent with Fig. 42.23?

42.21 • **CP** Silver has a Fermi energy of 5.48 eV. Calculate the electron contribution to the molar heat capacity at constant volume of silver, C_V, at 282 K. Express your result (a) as a multiple of R and (b) as a fraction of the actual value for silver, $C_V = 25.3\ \text{J/mol}\cdot\text{K}$. (c) Is the value of C_V due principally to the electrons? If not, to what is it due? (*Hint:* See Section 18.4.)

42.22 •• At the Fermi temperature T_F, $E_F = kT_F$ (see Exercise 42.18). When $T = T_F$, what is the probability that a state with energy $E = 2E_F$ is occupied?

42.23 •• For a solid metal having a Fermi energy of 8.480 eV, what is the probability, at room temperature, that a state having an energy of 8.540 eV is occupied by an electron?

Section 42.6 Semiconductors

42.24 • Pure germanium has a band gap of 0.67 eV. The Fermi energy is in the middle of the gap. (a) For temperatures of 255 K, 290 K, and 355 K, calculate the probability $f(E)$ that a state at the bottom of the conduction band is occupied. (b) For each temperature in part (a), calculate the probability that a state at the top of the valence band is empty.

42.25 • Germanium has a band gap of 0.67 eV. Doping with arsenic adds donor levels in the gap 0.01 eV below the bottom of the conduction band. At a temperature of 300 K, the probability is 4.4×10^{-4} that an electron state is occupied at the bottom of the conduction band. Where is the Fermi level relative to the conduction band in this case?

Section 42.7 Semiconductor Devices

42.26 •• (a) Suppose a piece of very pure germanium is to be used as a light detector by observing, through the absorption of photons, the increase in conductivity resulting from generation of electron–hole pairs. If each pair requires 0.67 eV of energy, what is the maximum wavelength that can be detected? In what portion of the spectrum does it lie? (b) What are the answers to part (a) if the material is silicon, with an energy requirement of 1.12 eV per pair, corresponding to the gap between valence and conduction bands in that element?

42.27 • **CP** At a temperature of 290 K, a certain *p-n* junction has a saturation current $I_S = 0.500$ mA. (a) Find the current at this temperature when the voltage is (i) 1.00 mV, (ii) −1.00 mV, (iii) 100 mV, and (iv) −100 mV. (b) Is there a region of applied voltage where the diode obeys Ohm's law?

42.28 • For a certain *p-n* junction diode, the saturation current at room temperature (20°C) is 0.950 mA. What is the resistance of this diode when the voltage across it is (a) 85.0 mV and (b) −50.0 mV?

42.29 •• (a) A forward-bias voltage of 15.0 mV produces a positive current of 9.25 mA through a *p-n* junction at 300 K. What does the positive current become if the forward-bias voltage is reduced to 10.0 mV? (b) For reverse-bias voltages of −15.0 mV and −10.0 mV, what is the reverse-bias negative current?

42.30 •• A *p-n* junction has a saturation current of 6.40 mA. (a) At a temperature of 300 K, what voltage is needed to produce a positive current of 40.0 mA? (b) For a voltage equal to the negative of the value calculated in part (a), what is the negative current?

42.31 • The saturation current for a junction diode is 10 microamps. Calculate the resistance of the diode at forward and reverse potential differences of 0.20 V when $T = 17°\text{C}$.

PROBLEMS

42.32 • When a diatomic molecule undergoes a transition from the $l = 2$ to the $l = 1$ rotational state, a photon with wavelength 54.3 μm is emitted. What is the moment of inertia of the molecule for an axis through its center of mass and perpendicular to the line connecting the nuclei?

42.33 •• **CP** (a) The equilibrium separation of the two nuclei in an NaCl molecule is 0.24 nm. If the molecule is modeled as charges $+e$ and $-e$ separated by 0.24 nm, what is the electric dipole moment of the molecule (see Section 21.7)? (b) The measured electric dipole moment of an NaCl molecule is $3.0 \times 10^{-29}\ \text{C}\cdot\text{m}$. If this dipole moment arises from point charges $+q$ and $-q$ separated by 0.24 nm, what is q? (c) A definition of the *fractional ionic character* of the bond is q/e. If the sodium atom has charge $+e$ and the chlorine atom has charge $-e$, the fractional ionic character would be equal to 1. What is the actual fractional ionic character for the bond in NaCl? (d) The equilibrium distance between nuclei in the hydrogen iodide (HI) molecule is 0.16 nm, and the measured electric dipole moment of the molecule is $1.5 \times 10^{-30}\ \text{C}\cdot\text{m}$. What is the fractional ionic character for the bond in HI? How does your answer compare to that for NaCl calculated in part (c)? Discuss reasons for the difference in these results.

42.34 • When a NaF molecule makes a transition from the $l = 3$ to the $l = 2$ rotational level with no change in vibrational quantum number or electronic state, a photon with wavelength 3.83 mm is emitted. A sodium atom has mass 3.82×10^{-26} kg, and a fluorine atom has mass 3.15×10^{-26} kg. Calculate the equilibrium separation between the nuclei in a NaF molecule. How does your answer compare with the value for NaCl given in Section 42.1? Is this result reasonable? Explain.

42.35 •• **CP** Consider a gas of diatomic molecules (moment of inertia I) at an absolute temperature T. If E_g is a ground-state energy and E_{ex} is the energy of an excited state, then the Maxwell–Boltzmann distribution (see Section 39.4) predicts that the ratio of the numbers of molecules in the two states is $n_{ex}/n_g = e^{-(E_{ex}-E_g)/kT}$. (a) Explain why the ratio of the number of molecules in the lth rotational energy *level* to the number of molecules in the ground-state ($l = 0$) rotational level is

$$\frac{n_l}{n_0} = (2l + 1)e^{-[l(l+1)\hbar^2]/2IkT}$$

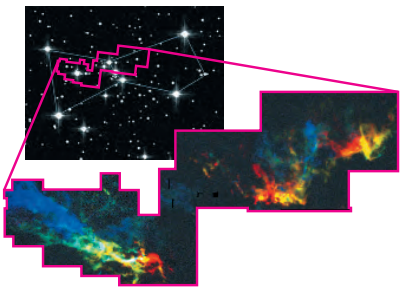
(*Hint:* For each value of l, how many states are there with different values of m_l?) (b) Determine the ratio n_l/n_0 for a gas of CO molecules at 300 K for (i) $l = 1$; (ii) $l = 2$; (iii) $l = 10$; (iv) $l = 20$; (v) $l = 50$. The moment of inertia of the CO molecule is given in Example 42.2 (Section 42.2). (c) Your results in part (b) show that as l is increased, the ratio n_l/n_0 first increases and then decreases. Explain why.

42.36 ••• **CALC** Part (a) of Problem 42.35 gives an equation for the number of diatomic molecules in the lth rotational level to the number in the ground-state rotational level. (a) Derive an expression for the value of l for which this ratio is the largest. (b) For the CO molecule at $T = 300$ K, for what value of l is this ratio a maximum? (The moment of inertia of the CO molecule is given in Example 42.2.)

42.37 •• Semiconductor devices can radiate heat up to a maximum power rating. If the device dissipates more power than that limit, it will heat up and explode, melt, or simply break down. A typical power rating for a diode is 500 mW. The power dissipated by any two-terminal device is the product of the current through it and the voltage across it. (a) Use Eq. (42.22) to estimate the power dissipated by a diode when it is forward biased at $V = 0.6$ V at a temperature of 300 K. The saturation current is 1.2×10^{-11} A. (b) Estimate to one significant figure the forward voltage at which the device will dissipate 500 mW. (c) Then what is the current?

42.38 •• Our galaxy contains numerous *molecular clouds,* regions many light-years in extent in which the density is high enough and the temperature low enough for atoms to form into molecules. Most of the molecules are H_2, but a small fraction of the molecules are carbon monoxide (CO). Such a molecular cloud in the constellation Orion is shown in **Fig. P42.38**. The upper image was made with an ordinary visible-light telescope; the lower image shows the molecular cloud in Orion as imaged with a radio telescope tuned to a wavelength emitted by CO in a rotational transition. The different colors in the radio image indicate regions of the cloud that are moving either toward us (blue) or away from us (red) relative to the motion of the cloud as a whole, as determined by the Doppler shift of the radiation. (Since a molecular cloud has about 10,000 hydrogen molecules for each CO molecule, it might seem more reasonable to tune a radio telescope to emissions from H_2 than to emissions from CO. Unfortunately, it turns out that the H_2 molecules in molecular clouds do not radiate in either the radio or visible portions of the electromagnetic spectrum.) (a) Using the data in Example 42.2 (Section 42.2), calculate the energy and wavelength of the photon emitted by a CO molecule in an $l = 1 \rightarrow l = 0$ rotational transition. (b) As a rule, molecules in a gas at temperature T will be found in a certain excited rotational energy level, provided the energy of that level is no higher than kT (see Problem 42.35). Use this rule to explain why astronomers can detect radiation from CO in molecular clouds even though the typical temperature of a molecular cloud is a very low 20 K.

Figure **P42.38**



42.39 • The force constant for the internuclear force in a hydrogen molecule (H_2) is $k' = 576$ N/m. A hydrogen atom has mass 1.67×10^{-27} kg. Calculate the zero-point vibrational energy for H_2 (that is, the vibrational energy the molecule has in the $n = 0$ ground vibrational level). How does this energy compare in magnitude with the H_2 bond energy of -4.48 eV?

42.40 • When an OH molecule undergoes a transition from the $n = 0$ to the $n = 1$ vibrational level, its internal vibrational energy increases by 0.463 eV. Calculate the frequency of vibration and the force constant for the interatomic force. (The mass of an oxygen atom is 2.66×10^{-26} kg, and the mass of a hydrogen atom is 1.67×10^{-27} kg.)

42.41 • The hydrogen iodide (HI) molecule has equilibrium separation 0.160 nm and vibrational frequency 6.93×10^{13} Hz. The mass of a hydrogen atom is 1.67×10^{-27} kg, and the mass of an iodine atom is 2.11×10^{-25} kg. (a) Calculate the moment of inertia of HI about a perpendicular axis through its center of mass. (b) Calculate the wavelength of the photon emitted in each of the following vibration-rotation transitions: (i) $n = 1, l = 1 \rightarrow n = 0, l = 0$; (ii) $n = 1, l = 2 \rightarrow n = 0, l = 1$; (iii) $n = 2, l = 2 \rightarrow n = 1, l = 3$.

42.42 • Suppose the hydrogen atom in HF (see the Bridging Problem for this chapter) is replaced by an atom of deuterium, an isotope of hydrogen with a mass of 3.34×10^{-27} kg. The force constant is determined by the electron configuration, so it is the same as for the normal HF molecule. (a) What is the vibrational frequency of this molecule? (b) What wavelength of light corresponds to the energy difference between the $n = 1$ and $n = 0$ levels? In what region of the spectrum does this wavelength lie?

42.43 •• Compute the Fermi energy of potassium by making the simple approximation that each atom contributes one free electron. The density of potassium is 851 kg/m^3, and the mass of a single potassium atom is 6.49×10^{-26} kg.

42.44 •• **CALC** The one-dimensional calculation of Example 42.4 (Section 42.3) can be extended to three dimensions. For the three-dimensional fcc NaCl lattice, the result for the potential energy of a pair of Na^+ and Cl^- ions due to the electrostatic interaction with all of the ions in the crystal is $U = -\alpha e^2/4\pi\epsilon_0 r$, where $\alpha = 1.75$ is the *Madelung constant*. Another contribution to the potential energy is a repulsive interaction at small ionic separation r due to overlap of the electron clouds. This contribution can be represented by A/r^8, where A is a positive constant, so the expression for the total potential energy is

$$U_{tot} = -\frac{\alpha e^2}{4\pi\epsilon_0 r} + \frac{A}{r^8}$$

(a) Let r_0 be the value of the ionic separation r for which U_{tot} is a minimum. Use this definition to find an equation that relates r_0 and A, and use this to write U_{tot} in terms of r_0. For NaCl, $r_0 = 0.281$ nm. Obtain a numerical value (in electron volts) of U_{tot} for NaCl. (b) The quantity $-U_{tot}$ is the energy required to remove an Na^+ ion and a Cl^- ion from the crystal. Forming a pair of neutral atoms from this pair of ions involves the release of 5.14 eV (the ionization energy of Na) and the expenditure of 3.61 eV (the electron affinity of Cl). Use the result of part (a) to calculate the energy required to remove a pair of neutral Na and Cl atoms from the crystal. The experimental value for this quantity is 6.39 eV; how well does your calculation agree?

42.45 ••• Metallic lithium has a bcc crystal structure. Each unit cell is a cube of side length $a = 0.35$ nm. (a) For a bcc lattice, what is the number of atoms per unit volume? Give your answer in terms of a. (*Hint:* How many atoms are there per unit cell?) (b) Use the result of part (a) to calculate the zero-temperature Fermi energy E_{F0} for metallic lithium. Assume there is one free electron per atom.

42.46 •• **DATA** To determine the equilibrium separation of the atoms in the HCl molecule, you measure the rotational spectrum of HCl. You find that the spectrum contains these wavelengths (among others): 60.4 μm, 69.0 μm, 80.4 μm, 96.4 μm, and 120.4 μm. (a) Use your measured wavelengths to find the moment of inertia of the HCl molecule about an axis through the center of mass and perpendicular to the line joining the two nuclei. (b) The value of l changes by ± 1 in rotational transitions. What value of l for the upper level of the transition gives rise to each of these wavelengths? (c) Use your result of part (a) to calculate the equilibrium separation of the atoms in the HCl molecule. The mass of a chlorine atom is 5.81×10^{-26} kg, and the mass of a hydrogen atom is 1.67×10^{-27} kg. (d) What is the longest-wavelength line in the rotational spectrum of HCl?

42.47 •• **DATA** The table gives the occupation probabilities $f(E)$ as a function of the energy E for a solid conductor at a fixed temperature T.

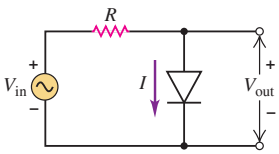
$f(E)$	0.064	0.173	0.390	0.661	0.856	0.950
E (eV)	3.0	2.5	2.0	1.5	1.0	0.5

To determine the Fermi energy of the solid material, you are asked to analyze this information in terms of the Fermi–Dirac distribution. (a) Graph the values in the table as E versus $\ln\{[1/f(E)] - 1\}$. Find the slope and y-intercept of the best-fit straight line for the data points when they are plotted this way. (b) Use your results of part (a) to calculate the temperature T and the Fermi energy of the material.

42.48 •• **DATA** A p-n junction is part of the control mechanism for a wind turbine that is used to generate electricity. The turbine has been malfunctioning, so you are running diagnostics. You can remotely change the bias voltage V applied to the junction and measure the current through the junction. With a forward-bias voltage of $+5.00$ mV, the current is $I_f = 0.407$ mA. With a reverse-bias voltage of -5.00 mV, the current is $I_r = -0.338$ mA. Assume that Eq. (42.22) accurately represents the current–voltage relationship for the junction, and use these two results to calculate the temperature T and saturation current I_S for the junction. [*Hint:* In your analysis, let $x = e^{eV/kT}$. Apply Eq. (42.22) to each measurement and obtain a quadratic equation for x.]

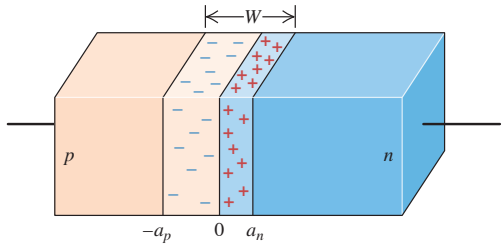
42.49 •• The circuit shown in **Fig. P42.49** is called a half-wave rectifier. The triangular symbol with the flat lower edge represents a diode, such that a forward bias drives the current downward. Use Eq. (42.22) to study the behavior of this circuit. Use a typical value for the saturation current: $I_S = 5 \times 10^{-13}$ A. Use a temperature of 300 K. (a) Any predicted current less than 1 μA can be considered a zero current. Determine the minimum voltage V_{out} for which the current I exceeds 1 μA. (b) The current increases dramatically for diode voltages that exceed that value. At a given diode voltage V_{out}, the diode has an effective resistance given by $R_D = V_{out}/I$. Use Eq. (42.22) to estimate the diode voltage $V_{out} = V_{max}$ for which $R_D = 100\ \Omega$. (c) What is the current I in that case? (d) If the resistance R is greater than 10 kΩ, then $R_D \ll R$ for larger currents, and there is a negligible additional voltage drop across the diode. This effectively pins $V_{out} = V_{max}$ for all input voltages $V_{in} > V_{max}$. With this perspective, estimate how this circuit would respond to a sinusoidal input voltage with amplitude 5 V.

Figure **P42.49**



42.50 ••• **CP CALC** A p-n junction includes p-type silicon, with donor atom density N_D, adjacent to n-type silicon, with acceptor atom density N_A. Near the junction, free electrons from the n side have diffused into the p side while free holes from the p side have diffused into the n side, leaving a "depletion region" of width W where there are no free charge carriers (**Fig. P42.50**). The width of the depletion region on the n side is a_n, and the width of the depletion region on the p side is a_p. Each depleted region has a constant charge density with magnitude equal to the corresponding donor atom density. Assume there is no net charge outside the depleted regions. Define an x-axis pointing from the p side toward the n side with the origin at the center of the junction. An electric field $\vec{E} = -E\hat{\imath}$ has developed in the depletion region, stabilizing the diffusion. (a) Use Gauss's law to determine the electric field on the p side of the junction, for $-a_p \le x \le 0$. (*Hint:* Use a cylindrical Gaussian surface parallel to the x-axis, with the left end outside the depletion region (where $\vec{E} = 0$) and the right side inside the region.) Note that the dielectric constant of silicon is $K = 11.7$. (b) Determine the electric field on the n side of the junction, for $0 \le x \le a_n$.

Figure **P42.50**



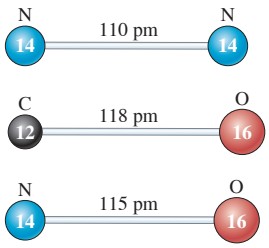
(c) Use continuity at $x = 0$ to determine a relationship between the depths a_p and a_n and in terms of N_A and N_D. (d) The electric potential $V(x)$ may be obtained by integrating the electric field, as shown in Eq. (23.18). Determine $V(x)$ in the region $-a_p \le x \le 0$ using the convention that $V(x) = 0$ for $x < -a_p$. (e) Similarly, determine $V(x)$ in the region $0 \le x \le a_n$. (f) What is the "barrier potential" $V_b = V(a_n) - V(-a_p)$? (g) The p side is doped with boron atoms with density $N_A = 1.00 \times 10^{16}\ \text{cm}^{-3}$. The n side is doped with arsenic atoms with density $N_D = 5.00 \times 10^{16}\ \text{cm}^{-3}$. The n side depletion depth is $a_n = 55.0$ nm. What is the p side depletion depth a_p? (h) What is the peak magnitude of the electric field, at $x = 0$? (i) What is the value of the barrier potential V_b?

42.51 ••• **CP** Consider the CO_2 molecule shown in Fig. 42.10c. The oxygen molecules have mass M_O and the carbon atom has mass M_C. Parameterize the positions of the left oxygen atom, the carbon atom, and the right oxygen atom using x_1, x_2, and x_3 as the respective rightward deviations from equilibrium. Treat the bonds as Hooke's-law springs with common spring constant k'. (a) Use Newton's second law to obtain expressions for $M_i\ddot{x}_i = M_i\, d^2x/dt^2$ in each case $i = 1, 2, 3$, where $M_{1,2,3} = (M_O, M_C, M_O)$. (*Note:* We represent time derivatives using dots.) Assume $X_C = 0$ at $t = 0$. (b) To ascertain the motion of the asymmetric stretching mode, set $x_1 = x_3 \equiv X_O$ and set $x_2 \equiv X_C$. Write the two independent equations that remain from your previous result. (c) Eliminate the sum $X_O + X_C$ from your equations. Use what remains to ascertain X_C in terms of X_O. (d) Substitute your expression for X_C into your equation for X_O to derive a harmonic oscillator equation $M_{eff}\ddot{X}_O = -k\dot{X}_O$. What is M_{eff}? (e) This equation has the solution $X_O(t) = A\cos(\omega t)$. What is the angular frequency ω? (f) Using the experimentally determined spring constant $k' = 1860$ N/m and the atomic masses $M_C = 12$ u and $M_O = 16$ u, where u $= 1.6605 \times 10^{-27}$ kg, to determine the oscillation frequency $f = \omega/2\pi$.

CHALLENGE PROBLEMS

42.52 ••• A colorless, odorless gas in a tank needs to be identified. The gas is nitrogen (N_2), carbon monoxide (CO), or nitric oxide (NO). (See **Fig. P42.52**.) White light is passed through a sample to obtain an absorption spectrum. Among others, the following wavelengths present dark bands on the spectrum: $\lambda = 4.2680\ \mu$m, $4.2753\ \mu$m, $4.2826\ \mu$m, $4.2972\ \mu$m, and $4.3046\ \mu$m. These represent transitions from the lowest rotational states. (a) What photon energies correspond to these wavelengths? Specify your answers to five significant figures. Note that the speed of light is 2.9979×10^8 m/s and $h = 4.1357 \times 10^{-15}$ eV · s. (b) Which one of the absorbed energies corresponds to transitions *from* an $l = 0$ state of the gas? (*Hint:* Use Eq. (42.9) along with the transition rules as a guide.) (c) Which one of the absorbed energies corresponds to a transition *to* an $l = 0$ state? (d) Determine the value of $\hbar\sqrt{k'/m_r}$, where k' is the effective spring constant of the molecule and m_r is its reduced mass. (*Hint:* Use the previous two results and Eq. (42.9).) (e) Determine the molecule's moment of inertia I. (f) The atomic masses and the bond

Figure P42.52



lengths for the three molecule options are shown in Fig. P42.52, where $u = 1.6605 \times 10^{-27}$ kg is one atomic mass unit. Based on the data, what is the mystery gas? (g) What is the reduced mass m_r in terms of u? (h) What is the spring constant k'?

42.53 ••• **CALC** Consider a system of N free electrons within a volume V. Even at absolute zero, such a system exerts a pressure p on its surroundings due to the motion of the electrons. To calculate this pressure, imagine that the volume increases by a small amount dV. The electrons will do an amount of work $p\,dV$ on their surroundings, which means that the total energy E_{tot} of the electrons will change by an amount $dE_{tot} = -p\,dV$. Hence $p = -dE_{tot}/dV$. (a) Show that the pressure of the electrons at absolute zero is

$$p = \frac{3^{2/3}\pi^{4/3}\hbar^2}{5m}\left(\frac{N}{V}\right)^{5/3}$$

(b) Evaluate this pressure for copper, which has a free-electron concentration of $8.45 \times 10^{28}\ \text{m}^{-3}$. Express your result in pascals and in atmospheres. (c) The pressure you found in part (b) is extremely high. Why, then, don't the electrons in a piece of copper simply explode out of the metal?

42.54 ••• **CALC** When the pressure p on a material increases by an amount Δp, the volume of the material will change from V to $V + \Delta V$, where ΔV is negative. The *bulk modulus* B of the material is defined to be the ratio of the pressure change Δp to the absolute value $|\Delta V/V|$ of the fractional volume change. The greater the bulk modulus, the greater the pressure increase required for a given fractional volume change, and the more incompressible the material (see Section 11.4). Since $\Delta V < 0$, the bulk modulus can be written as $B = -\Delta p/(\Delta V/V_0)$. In the limit that the pressure and volume changes are very small, this becomes

$$B = -V\frac{dp}{dV}$$

(a) Use the result of Challenge Problem 42.53 to show that the bulk modulus for a system of N free electrons in a volume V at low temperatures is $B = \frac{5}{3}p$. (*Hint:* The quantity p in the expression $B = -V(dp/dV)$ is the *external* pressure on the system. Can you explain why this is equal to the *internal* pressure of the system itself, as found in Challenge Problem 42.53?) (b) Evaluate the bulk modulus for the electrons in copper, which has a free-electron concentration of $8.45 \times 10^{28}\ \text{m}^{-3}$. Express your result in pascals. (c) The actual bulk modulus of copper is 1.4×10^{11} Pa. Based on your result in part (b), what fraction of this is due to the free electrons in copper? (This result shows that the free electrons in a metal play a major role in making the metal resistant to compression.) What do you think is responsible for the remaining fraction of the bulk modulus?

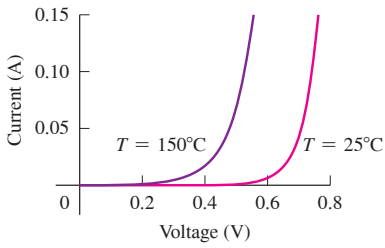
42.55 •• In the discussion of free electrons in Section 42.5, we assumed that we could ignore the effects of relativity. This is not a safe assumption if the Fermi energy is greater than about $\frac{1}{100}mc^2$ (that is, more than about 1% of the rest energy of an electron). (a) Assume that the Fermi energy at absolute zero, as given by Eq. (42.19), is equal to $\frac{1}{100}mc^2$. Show that the electron concentration is

$$\frac{N}{V} = \frac{2^{3/2}m^3c^3}{3000\pi^2\hbar^3}$$

and determine the numerical value of N/V. (b) Is it a good approximation to ignore relativistic effects for electrons in a metal such as copper, for which the electron concentration is $8.45 \times 10^{28}\ \text{m}^{-3}$? Explain. (c) A *white dwarf star* is what is left behind by a star like the sun after it has ceased to produce energy by nuclear reactions. (Our own sun will become a white dwarf star in another 8×10^9 years or so.) A typical white dwarf has mass 2×10^{30} kg (comparable to the sun) and radius 6000 km (comparable to that of the earth). The gravitational attraction of different parts of the white dwarf for each other tends to compress the star; what prevents it from compressing is the pressure of free electrons within the star (see Challenge Problem 42.53). Use both of the following assumptions to estimate the electron concentration within a typical white dwarf star: (i) the white dwarf star is made of carbon, which has a mass per atom of 1.99×10^{-26} kg; and (ii) all six of the electrons from each carbon atom are able to move freely throughout the star. (d) Is it a good approximation to ignore relativistic effects in the structure of a white dwarf star? Explain.

MCAT-STYLE PASSAGE PROBLEMS

Diode Temperature Sensor. The current–voltage characteristics of a forward-biased *p-n* junction diode depend strongly on temperature, as shown in the figure. As a result, diodes can be used as temperature sensors. In actual operation, the voltage is adjusted to keep the current through the diode constant at a specified value, such as 100 mA, and the temperature is determined from a measurement of the voltage at that current.



42.56 The sensitivity of a diode thermometer depends on how much the voltage changes for a given temperature change, with the current remaining constant. What is the sensitivity for this diode thermometer, operated at 100 mA, for a temperature change from 25°C to 150°C? (a) +0.2 mV/°C; (b) +2.0 mV/°C; (c) −0.2 mV/°C; (d) −2.0 mV/°C.

42.57 Which statement best explains the temperature dependence of the current–voltage characteristics that the graph shows? At higher temperatures: (a) The band gap is larger, so the electron–hole pairs have more energy, which causes the current at a given voltage to be larger. (b) More electrons can move to the conduction band, which causes the current at a given voltage to be larger. (c) All of the electrons in the valence band move to the conduction band, and the diode behaves like a metal and follows Ohm's law. (d) The acceptor and donor impurity atoms are free to move through the material, which causes the current at a given voltage to be larger.

42.58 If the voltage rather than the current is kept constant, what happens as the temperature increases from 25°C to 150°C? (a) At first the current increases, then it decreases. (b) The current increases. (c) The current decreases, eventually approaching zero. (d) The current does not change unless the voltage also changes.

ANSWERS

Chapter Opening Question ?

(i) Venus must radiate energy into space at the same rate that it receives energy in the form of sunlight. However, carbon dioxide (CO_2) molecules in the atmosphere absorb infrared radiation emitted by the surface of Venus and re-emit it toward the ground. This involves a transition between vibrational states of the CO_2 molecule (see Section 42.2). To compensate for this and to maintain the balance between emitted and received energy, the surface temperature of Venus and hence the rate of blackbody radiation from the surface increase.

Key Example VARIATION Problems

VP42.3.1 (a) 1.240×10^{-26} kg (b) 1.651×10^{-46} kg · m^2
(c) 0, 0.4203 meV, 1.261 meV
VP42.3.2 (a) 2.64×10^{-47} kg · m^2 (b) 1.627×10^{-27} kg (c) 0.127 nm
VP42.3.3 (a) 3.853×10^{-46} kg · m^2 (b) 1.690×10^{-26} kg
(c) 0.1510 nm
VP42.3.4 (a) $f = 6.538 \times 10^{13}$ Hz, $\lambda = 4.585\,\mu$m
(b) $f = 6.527 \times 10^{13}$ Hz, $\lambda = 4.593\,\mu$m
VP42.7.1 (a) $E_F + 0.71kT$ (b) $E_F - 2.2kT$
VP42.7.2 (a) 0.729 (b) 0.170 (c) 0.0187
VP42.7.3 (a) 8.8×10^{-19} J $= 5.5$ eV (b) 1.4×10^6 m/s
(c) 9.9×10^{40} states/J $= 1.6 \times 10^{22}$ states/eV
VP42.7.4 (a) 5.4 eV (b) 5.7 eV
VP42.9.1 (a) 3.79×10^{-5} (b) 5.46×10^{-14}
VP42.9.2 (a) 1.65 (b) 2.72
VP42.9.3 (a) 0.329 eV (b) 0.351 eV
VP42.9.4 (a) 0.0405 (b) 0.0643

Bridging Problem

(a) 0.278 eV
(b) 1.74×10^{-25}